

Can He

hecanhappy@gmail.com | (617) 513-2316 | Boston, MA | bostonsuprememantou.github.io

EDUCATION

Northeastern University - Boston Campus , Boston, MA <i>M.S. in Computer Science, GPA: 3.94 / 4.0</i> Coursework: Computer Systems, Algorithms, Object-Oriented Design, Discrete Structures, Foundations of AI, Machine Learning	Sep 2023 - Apr 2026
Northeastern University - Boston Campus , Boston, MA <i>M.S. in Project Management, GPA: 3.761 / 4.0</i>	Sep 2013 - Jan 2016
Chongqing University of Technology , Chongqing, China <i>B.E. in Mechanical Engineering</i>	Sep 2008 - Jun 2012

TECHNICAL SKILLS

Languages: Python, Java, C, R, MQL4, SQL
AI / Computer Vision: OpenCV, PyTorch, Ultralytics YOLO, Scikit-Learn, NumPy, Pandas, image preprocessing, postprocessing
Backend / Data: RabbitMQ, AWS S3, SQLAlchemy, Alembic, pytest, DAOs, async upload queues, traceability metadata
DevOps / Observability: Docker, Kubernetes, Grafana, Prometheus, systemd, Git, CI-ready testing workflows, runtime health checks

WORK EXPERIENCE

Tristar AI , Boston, MA <i>Junior Software Developer</i> - Own end-to-end production computer vision pipelines for manufacturing customers, converting inspection requirements into camera capture, inference, traceability, PLC feedback, and deployment workflows. - Design and maintain Python services for image preprocessing/postprocessing, task/result schemas, RabbitMQ/AWS S3 publishing, PLC rejection logic, runtime health checks, and deployment configuration. - Harden deployments with pytest coverage, test doubles, monitoring hooks, Grafana/Prometheus metrics, Docker/Kubernetes packaging, and operator runbooks.	May 2026 - Present
Tristar AI , Boston, MA <i>Software Development Intern</i> - Independently implemented data-access objects, upload schedulers, camera/data capture utilities, defect-analysis scripts, configuration loaders, and local test flows for customer-facing AI systems. - Built validation coverage across camera, messaging, storage, and inference boundaries using pytest, dependency injection, local fixtures, and CI-ready project structure.	Jun 2025 - Jan 2026
Northeastern University , Boston, MA <i>Teaching Assistant - DS5220 Supervised Machine Learning</i> Supported ML course delivery by grading assignments, preparing answer keys, explaining supervised-learning concepts, and leading office hours and review sessions.	Sep 2024 - Jun 2025

SELECTED PROJECTS

Medegen Manufacturing Defect Detection Pipeline

Python, OpenCV, PyTorch, Ultralytics YOLO, RabbitMQ, AWS S3, PLC, systemd

- Owned all tub defect detection work for four defect categories, spanning image preprocessing, model inference, postprocessing, classification, and line/camera-specific filtering.
- Implemented and integrated capture, inference, rejection, results, station-status, alerting, and outbox modules in a six-camera, three-line production system with multiprocessing workers.
- Delivered real-time quality feedback through FIFO-triggered capture, PLC-driven rejection, RabbitMQ/AWS S3 publishing, Slack alerts, auditable records, and runtime failure-mode tests.

Amcor Bottle Traceability Pipeline

Python, OpenCV, Terpsichore, AWS S3, RabbitMQ, Prometheus

- Built the full traceability pipeline connecting camera frames, station/team/factory metadata, task lifecycle states, result payloads, asynchronous upload queues, and monitored execution.

Greif Defect Data Analysis Pipeline

Python, Pandas, OpenCV, sensor/data streams

- Performed end-to-end defect analysis across records, image evidence, sensor/process streams, and timing signals to identify recurring failure patterns and support downstream inspection workflows.

Quantitative Trading & RL Systems

Python, MQL4, backtesting, MT4 gateway, reinforcement learning

- Built XAUUSD/FX research systems with spread/slippage/commission-aware backtests, equity curves, risk metrics, trade ledgers, MT4 command gateways, optimization reports, and RL evaluation.

EARLIER ENGINEERING EXPERIENCE

Wuhu Automobile Advanced Technology Institute Co., Ltd. , Wuhu, Anhui, China <i>Project Manager</i> Led a 23-person engineering team developing lightweight carbon-fiber automotive components, achieving a 42% weight reduction on targeted Chery Tiggo 8 Pro PHEV components while managing testing, cost control, and production integration.	Sep 2020 - Sep 2023
Chery Holding Group Co., Ltd. , Wuhu, Anhui, China <i>Project Manager</i> Optimized chassis adaptation for fuel and hybrid vehicle lines through testing, model adjustment, technical data management, QC collaboration, patent support, and production guidance.	Sep 2016 - Sep 2020